

ALIGARH INSTITUTE OF TECHNOLOGY

Project Exhibition — Final Documentation Report

ClinicFlow Pro

Modern Clinic Management SaaS for Pakistan's Healthcare Sector

Live Project: clinic-flow-landingpage.vercel.app

Project Lead / Group Leader	Abdul Hannan Ghaznavi
Team Member	Hasnat Ahmed
Lead Role	Backend Architecture, Database Design & System Logic
Tech Stack	PHP 8.0+, Python 3.x, MySQL, HTML5, CSS3, Tailwind CSS, JavaScript, GSAP
Category	Healthcare Management SaaS Platform

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1. Executive Summary

ClinicFlow Pro is a complete, all-in-one clinic management platform built to digitize day-to-day operations for clinics, hospitals, and individual practitioners across Pakistan. The system replaces fragmented paper-based workflows — appointment registers, handwritten prescriptions, and manual patient files — with a single web-based platform accessible to administrators, doctors, receptionists, and patients.

The platform currently includes 42+ functional modules spanning appointment scheduling, digital prescriptions, patient timelines, billing, analytics, AI-assisted diagnosis, and communication tools such as WhatsApp reminders and an in-app chatbot. It was designed with the local healthcare market in mind, including PKR-based pricing tiers and considerations for Urdu-speaking patients.

This documentation covers the problem the project addresses, the reasoning behind our design decisions, the complete technical implementation, and the roadmap for future development.

2. Problem Statement

The majority of small and mid-sized clinics in Pakistan still rely heavily on paper-based systems for patient records, appointment booking, and prescriptions. Through research and observation of local clinic operations, the following recurring issues were identified:

- Patient records are kept on paper registers or scattered spreadsheets, making history lookups slow and error-prone.
- Appointments are managed manually over phone calls, leading to double-bookings and missed slots.
- Handwritten prescriptions are often illegible, causing confusion at pharmacies and risk to patients.
- Clinic staff spend hours each week on paperwork instead of focusing on patient care.
- There is no easy way for patients to view their own medical history, upcoming appointments, or prescriptions.
- Clinic owners have no analytics or insight into revenue, doctor performance, or patient flow.
- Existing international SaaS solutions are priced in USD and are not adapted for local payment methods, language, or pricing expectations.

These problems directly translate into lost time, lost revenue, and a lower standard of patient care — especially for clinics that cannot afford expensive enterprise hospital management systems.

3. Our Idea & Approach

As the project lead, my starting point was simple: most hospital management systems available either target large hospitals with enterprise budgets, or are too generic and not adapted to how Pakistani clinics actually operate — in terms of language, pricing, and communication channels (most patients and staff communicate via WhatsApp).

The idea behind ClinicFlow Pro was to build a lightweight but feature-complete clinic management SaaS that:

- Is affordable enough for a single-doctor clinic, but scalable enough for a multi-branch setup.
- Provides separate, role-based dashboards so every user — admin, doctor, receptionist, and patient — only sees what is relevant to them.
- Uses WhatsApp as a first-class communication channel for reminders and prescriptions, since that is how clinics already talk to patients.
- Integrates AI to assist with symptom checking, differential diagnosis suggestions, and patient-facing health queries — reducing the burden on front-desk staff.
- Is priced in PKR with tiers that match what local clinics can realistically afford, rather than copying international SaaS pricing models.

The approach was to design the database and core modules first (patients, doctors, appointments, prescriptions), validate the workflow logic, and then layer on advanced features such as analytics, AI assistance, and the patient portal once the core system was stable.

4. Solution Overview

4.1 Core Workflow

1. A patient is registered into the system either by the receptionist or via self-registration through the patient portal.
2. Appointments are booked against doctor availability, with real-time conflict detection to prevent double-booking.
3. During consultation, the doctor records diagnosis notes and issues a digital prescription, which is instantly available to the patient.
4. The patient timeline is automatically updated with every visit, diagnosis, prescription, and test result, building a complete chronological medical history.
5. Automated WhatsApp/SMS reminders are sent for upcoming appointments and prescription refills.
6. Admins use the analytics dashboard to monitor clinic revenue, appointment volume, and doctor performance in real time.

4.2 What Makes This Solution Different

- Built specifically for the Pakistani market — PKR pricing, Urdu-friendly design considerations, WhatsApp-first communication.
- AI Symptom Checker and AI Diagnosis Assistant give doctors and patients a head start before/after a consultation.
- A free tier exists, lowering the barrier to entry for small clinics that cannot commit to a paid plan immediately.
- Drag-and-drop customizable dashboards let each clinic adapt the interface to their own workflow.

5. Key Features & Modules

ClinicFlow Pro is built around 15 major feature groups, which together expand into 42+ individual modules across the four dashboards. The table below summarizes each major feature area.

Feature	Description
User Roles	4 distinct roles — Admin, Doctor, Receptionist, and Patient — each with a tailored dashboard and permission set.
Smart Scheduling	Real-time doctor availability, automatic conflict detection, and automated appointment reminders.
Patient Timeline	A complete, chronological record of every visit, diagnosis, and treatment for each patient.
Digital Prescriptions	E-prescriptions with dosage tracking and QR codes that pharmacies can scan to verify authenticity.
AI Symptom Checker	Patients describe symptoms and receive an intelligent analysis with urgency indicators.
Analytics Dashboard	Real-time insights into revenue, patient flow, and doctor performance for clinic admins.
Doctor-Patient Chat	Secure, in-app messaging between doctors and their patients.
Health Predictor	AI-powered risk assessment based on a patient's medical history.
Bank-Grade Security	Role-based access control, session management, and full audit trails for every action.
Subscription Plans	A free tier plus upgradeable Pro plans for clinics that need unlimited features.
AI Health Assistant	A 24/7 chatbot capable of handling 500+ types of patient queries.
WhatsApp Integration	Automated appointment reminders and prescription delivery via WhatsApp.
Drag & Drop Dashboard	Each user can rearrange and save their own dashboard layout.
AI Diagnosis Assistant	AI-powered differential diagnosis suggestions based on reported symptoms.

6. System Architecture

ClinicFlow Pro follows a traditional but well-organized server-rendered architecture, optimized for reliability and ease of deployment on shared hosting environments commonly available in Pakistan.

6.1 High-Level Architecture

- **Presentation Layer:** HTML5, Tailwind CSS, and vanilla JavaScript power the UI, with GSAP used for smooth animations and transitions across the landing page and dashboards.
- **Application Layer:** PHP 8.0+ handles all server-side logic, including authentication, role-based routing, business rules, and API endpoints consumed by the frontend via AJAX.
- **Data Layer:** MySQL stores all persistent data — patients, doctors, appointments, prescriptions, billing, and audit logs — accessed through prepared statements to prevent SQL injection.
- **AI Layer:** External AI APIs (via Python 3.x) power the Symptom Checker, AI Health Assistant chatbot, Health Predictor, and Diagnosis Assistant, called from the backend.
- **Communication Layer:** WhatsApp Business API (or a gateway integration) and SMS gateways are used for appointment reminders and prescription delivery.

6.2 Deployment

The application is served over Apache with HTTPS enforced across all routes. The current production build is hosted with a live demo landing page available for evaluation, while the application backend runs on a PHP/MySQL stack with Python for AI services, suitable for standard LAMP hosting.

7. Database Design

The system is built on a normalized MySQL schema with 8 core tables (with additional supporting tables for chat, notifications, and AI interaction logs as features were expanded). The core tables are summarized below.

Table	Purpose
users	Stores login credentials and role assignment (Admin, Doctor, Receptionist, Patient).
patients	Patient profile information, contact details, and medical history references.
doctors	Doctor profiles, specializations, and availability schedules.
appointments	Appointment bookings with date, time, status, and conflict-checking fields.
prescriptions	Digital prescriptions linked to a patient and doctor, with dosage and QR code data.
patient_timeline	Chronological log of visits, diagnoses, and treatments for each patient.
billing	Invoice and payment records linked to appointments and subscription plans.
audit_logs	Security and activity logs tracking user actions across the system.

Each table uses primary/foreign key relationships to maintain referential integrity — for example, every appointment links to a patient_id and doctor_id, and every prescription links back to the appointment it was issued from. This design ensures the patient timeline can be reconstructed accurately from related records at any time.

8. Technology Stack

The stack was chosen for reliability, wide hosting availability in Pakistan, and a balance between development speed and performance.

Layer	Technologies Used
Frontend	HTML5, CSS3, Tailwind CSS, JavaScript (ES6+), GSAP
Backend	PHP 8.0+, Python 3.x
Database	MySQL
Server	Apache (LAMP stack)
Security	HTTPS, Role-Based Access Control (RBAC), Audit Logs
AI Integration	AI APIs for Symptom Checker, Diagnosis Assistant, Health Predictor, and Chatbot
Communication	WhatsApp Integration, SMS Gateway

9. User Roles & Dashboards

The platform provides four dashboards, each tailored to the responsibilities of that role, totaling 42+ individual pages across the system.

9.1 Admin Dashboard

- Overview of total patients, doctors, appointments, and clinic uptime/performance.
- Access to the full analytics dashboard — revenue tracking, doctor performance, patient flow.
- Manage staff accounts, roles, and permissions.
- View system-wide audit logs.

9.2 Doctor Dashboard

- View daily schedule and patient queue with real-time availability status.
- Access full patient timeline and history before each consultation.
- Issue digital prescriptions with QR codes for pharmacy verification.
- Use the AI Diagnosis Assistant for differential diagnosis suggestions.
- Communicate with patients via secure in-app chat.

9.3 Receptionist Dashboard

- Register new patients and manage existing patient records.
- Book, reschedule, and cancel appointments with conflict detection.
- Handle billing and payment records.

9.4 Patient Portal

- View upcoming appointments and consultation history (e.g., consultation, blood test status tracking).
- Access digital prescriptions and download/print as needed.
- Use the AI Symptom Checker before booking an appointment.
- Receive WhatsApp/SMS reminders for appointments and medication.

10. AI-Powered Capabilities

AI integration was one of the core differentiators of this project, aimed at reducing the workload on clinic staff and giving patients more confidence before and after consultations.

10.1 AI Symptom Checker

Patients can describe their symptoms in plain language and receive an intelligent analysis that includes urgency indicators — helping them decide whether a condition needs immediate attention or can wait for a scheduled appointment.

10.2 AI Diagnosis Assistant

For doctors, the AI Diagnosis Assistant analyzes reported symptoms and patient history to suggest a list of possible differential diagnoses, which the doctor can use as a starting point during consultation — not as a replacement for clinical judgment.

10.3 Health Predictor

Using a patient's recorded medical history, the Health Predictor module performs an AI-powered risk assessment, flagging potential long-term health risks that may warrant preventive care.

10.4 AI Health Assistant (Chatbot)

A 24/7 chatbot is available to patients for general queries — covering over 500 distinct response scenarios, including appointment-related questions, general health information, and platform usage help.

11. Security & Compliance

Given that the system handles sensitive medical data, security was treated as a core requirement rather than an afterthought.

- HTTPS enforced across the entire application to encrypt data in transit.
- Role-Based Access Control (RBAC) ensures each user can only access data and functions relevant to their role.
- Session management with secure session handling and timeout policies.
- Full audit trails log every significant action taken within the system, supporting accountability and traceability.
- The platform is designed with HIPAA-aligned principles in mind for handling patient data responsibly.

12. Project Timeline

The project was developed in 8 sequential phases, moving from core backend logic outward to advanced AI features and final presentation.

Phase	Activities
Phase 1 — Research & Planning	Studied common pain points in local clinics, defined user roles, and designed the database schema and core entity relationships.
Phase 2 — Core Backend Development	Built authentication, role-based access, and the foundational patient, doctor, and appointment modules in PHP/MySQL.
Phase 3 — Dashboard & UI Development	Designed and built all 4 dashboards (Admin, Doctor, Receptionist, Patient) — 42+ pages in total — using Tailwind CSS and GSAP animations.
Phase 4 — Advanced Features	Implemented digital prescriptions with QR codes, patient timeline, analytics dashboard, and billing/subscription system.
Phase 5 — AI Integration	Integrated the AI Symptom Checker, AI Diagnosis Assistant, Health Predictor, and the AI Health Assistant chatbot.
Phase 6 — Communication	Added WhatsApp/SMS reminder integration.
Phase 7 — Security Hardening & Testing	Implemented RBAC, audit logging, HTTPS enforcement, and conducted end-to-end testing across all roles.
Phase 8 — Landing Page & Deployment	Built the public landing page with live stats, pricing poll, testimonials, and deployed the project for demonstration.

13. Team & Contribution

This project was developed by a two-person team, with clearly divided responsibilities to ensure both the backend system and the frontend experience reached a professional standard.

13.1 Abdul Hannan Ghaznavi — Group Leader & Lead Full Stack Developer

- Architected the complete server-side logic, database schema (8 core tables), and security infrastructure.
- Designed and implemented 42+ functional modules covering appointments, prescriptions, billing, AI integration, and analytics.
- Led overall project planning, system design decisions, and coordinated feature scope across approximately 250+ development hours.
- Responsible for integrating AI APIs, WhatsApp/SMS communication, and implementing the security and audit logging layer.

13.2 Hasnat Ahmed — Lead Frontend Developer

- Designed and built all 4 dashboards and 42+ pages, including the cinematic landing page using Tailwind CSS and GSAP.
- Focused on UI/UX consistency across roles, ensuring each dashboard felt purpose-built for its user type.
- Contributed approximately 220+ hours to frontend design, animation, and responsive layout work.

Together, the team combined backend system design with a polished, animated frontend to deliver a product that feels production-ready rather than a typical academic prototype.

14. Pricing Strategy

Rather than guessing at a price point, the landing page includes a live pricing poll that lets visiting clinics vote on what they consider fair pricing for the platform — directly informing our go-to-market pricing strategy. The proposed PKR-based tiers are outlined below.

Tier	Description
Free Tier	Basic features for individual practitioners or very small clinics to get started with no upfront cost.
Rs 5,000 – 10,000 / month	Basic paid tier — suitable for small clinics needing core scheduling and prescription features.
Rs 10,000 – 20,000 / month	Standard tier — adds analytics and extended patient management features.
Rs 20,000 – 30,000 / month	Pro tier — includes AI features (Symptom Checker, Diagnosis Assistant).
Rs 30,000 – 50,000 / month	Enterprise tier — multi-branch support and advanced analytics.
Rs 50,000 – 100,000 / month	Premium tier with dedicated support.

This approach ensures pricing reflects what the local market is actually willing to pay, rather than being based on assumptions or international SaaS benchmarks.

15. Challenges Faced

- Designing a single database schema flexible enough to support 4 very different user roles without creating redundant or conflicting data structures.
- Implementing real-time appointment conflict detection that works reliably across concurrent bookings from receptionists and patients.
- Integrating multiple AI features (Symptom Checker, Diagnosis Assistant, Health Predictor, Chatbot) while keeping response times acceptable for a clinical setting.
- Building 42+ pages across 4 dashboards while maintaining a consistent design language and animation style throughout.
- Balancing feature richness with simplicity — ensuring the system remains usable for clinic staff with limited technical background.
- Implementing a security and audit logging layer thorough enough to be credible for handling medical data, within the scope of a student project.

16. Future Scope

- Multi-branch / multi-clinic support for healthcare groups operating several locations under one account.
- Native mobile applications (Android/iOS).
- Deeper integration with local pharmacy networks for automatic prescription fulfillment via QR codes.
- Insurance claim processing and integration with local insurance providers.
- Expanded Urdu-language support across the entire interface, including AI chatbot responses.
- Telemedicine module for full video consultations directly within the platform.
- More granular analytics, including predictive scheduling based on historical patient flow data.

17. Conclusion

ClinicFlow Pro demonstrates how a thoughtfully designed, locally-aware SaaS product can address real operational problems faced by clinics in Pakistan — without requiring the budget or complexity of enterprise hospital management systems.

By combining a solid PHP/MySQL/Python backend, a polished, animated frontend, AI-assisted tools, and communication channels that match how local clinics already operate (such as WhatsApp), the project goes beyond a typical academic exercise and reflects a genuinely deployable product.

As the group leader, my goal throughout this project was to ensure that every module — from the database schema to the AI integrations — was built with production quality in mind, while also creating a learning experience that strengthened both backend architecture and frontend collaboration skills across the team.

We welcome feedback from visitors and evaluators on the following pages, which will directly inform the next iteration of ClinicFlow Pro.

18. Visitor Feedback

Page 1 of 4 — Please share your thoughts on the ClinicFlow Pro project. Your feedback helps us improve future versions of the platform.

#	Visitor Name	Comments / Feedback	Signature
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18. Visitor Feedback

Page 2 of 4 — Please share your thoughts on the ClinicFlow Pro project. Your feedback helps us improve future versions of the platform.

#	Visitor Name	Comments / Feedback	Signature
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18. Visitor Feedback

Page 3 of 4 — Please share your thoughts on the ClinicFlow Pro project. Your feedback helps us improve future versions of the platform.

#	Visitor Name	Comments / Feedback	Signature
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18. Visitor Feedback

Page 4 of 4 — Please share your thoughts on the ClinicFlow Pro project. Your feedback helps us improve future versions of the platform.

#	Visitor Name	Comments / Feedback	Signature
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